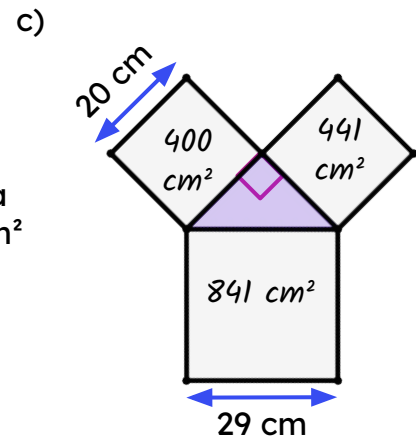
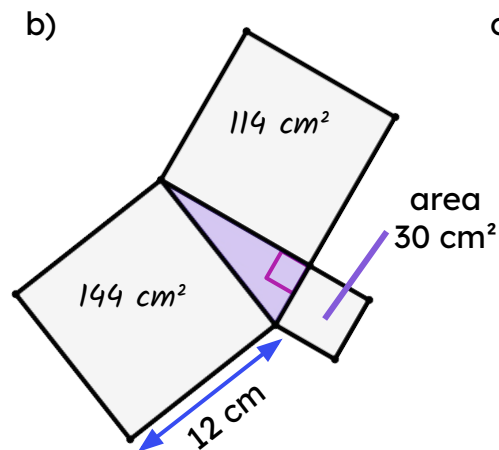
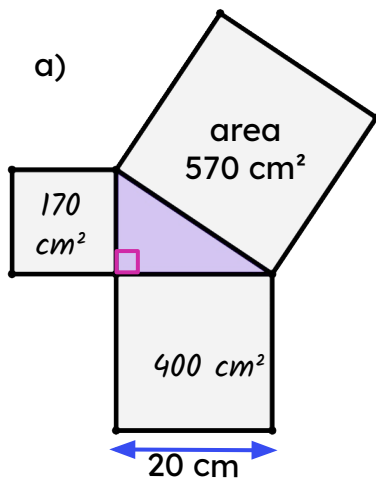




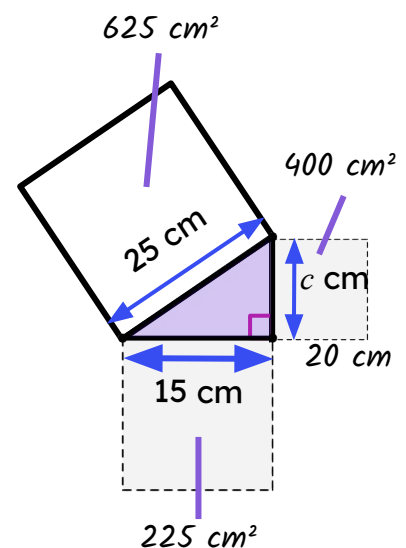
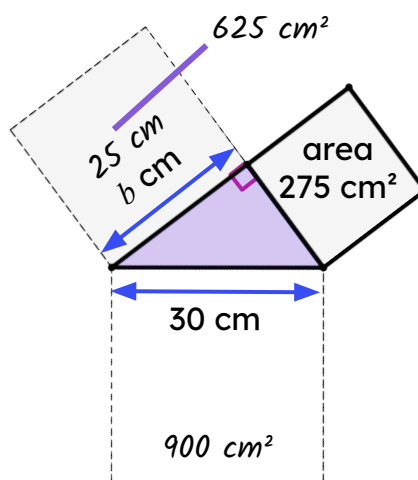
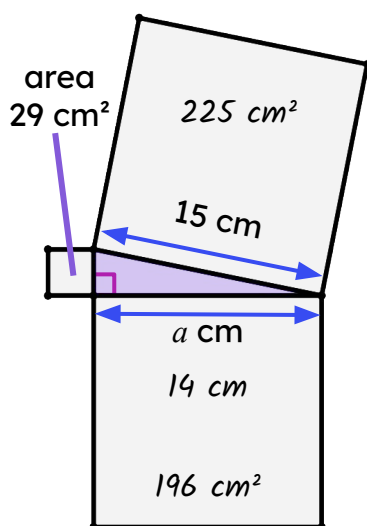
Length of a shorter side

Task A: Squares from the sides of a triangle

1) Find the area of each unlabelled square.



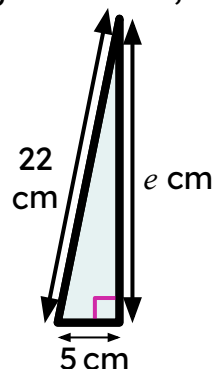
2) By drawing on any missing squares first and calculating their areas, find the length of each side labelled with a letter.



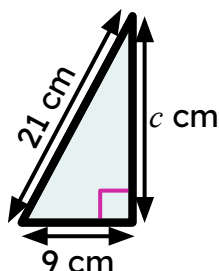
Task B: Using Pythagoras' theorem to find a shorter side

1) Fill in all the blanks to complete the method to calculate the length of side e , rounded to 1 decimal place.

$$\begin{aligned} e^2 + 5^2 &= 22^2 \\ e^2 + 25 &= 484 \\ e^2 &= 484 - 25 \\ e^2 &= 459 \\ e &= \sqrt{459} = 21.4 \text{ (1 d.p.)} \end{aligned}$$



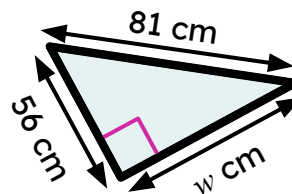
2) Which of these are correct representations of Pythagoras' theorem to find the length of side c ?



- ☐ a $c^2 = 21^2 + 9^2$
☒ b $c^2 = 21^2 - 9^2$
☒ c $21^2 = 9^2 + c^2$
☒ d $21^2 = c^2 + 9^2$
☐ e $21^2 = 9^2 - c^2$
☐ f $21^2 = c^2 - 9^2$

3) Using Pythagoras' theorem, show fully that the value of w cm is 58.52 cm when rounded to 2 decimal places.

$$\begin{aligned} 56^2 + w^2 &= 81^2 \\ w^2 &= 81^2 - 56^2 \\ w^2 &= 3425 \\ w &= \sqrt{3425} = 58.5234996 \\ w &= 58.52, \text{ after rounding.} \end{aligned}$$

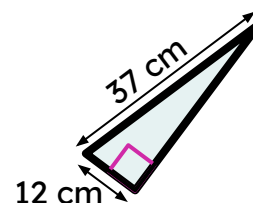


4) Show that the perimeter of this triangle is 84 cm.

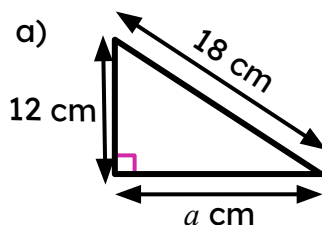
Let the missing length be called x .

$$\begin{aligned} x^2 &= 37^2 - 12^2 = 1225 \\ x &= \sqrt{1225} = 35 \end{aligned}$$

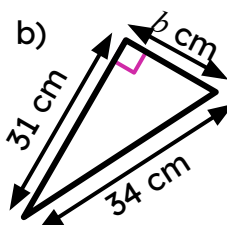
Perimeter = 12 cm + 37 cm + 35 cm
 Perimeter = 84 cm



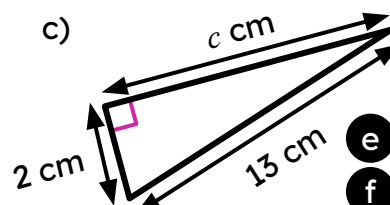
5) Match each triangle to a calculation and answer to find the length of its missing side.



g
i



d
h



e j
f

calculation ☒ d $34^2 - 31^2$ ☐ e $169 - 2^2$ ☐ f 165 ☐ g $324 - 144$

length of missing side ☒ h 13.96 cm ☒ i 13.42 cm ☒ j 12.84 cm

